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Monitoring for Surface Geological Deformations in Layla-Alaflaj in Central Saudi Arabia Using Satellite Based Remote Sensing and Differential Interferometry (DInSAR) Techniques

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Abstract

Differential Interferometry Synthetic Aperture Radar (DInSAR) is a valuable remote sensing technique that enables monitoring of large geographical areas at a low cost. One of the main applications for this technique is its ability to detect and monitor terrain deformations and morphological changes through quantitative interpretation of measured surface displacements. This is especially advantageous for deformations that are gradually occurring over the years. Layla-Alaflaj is in central Saudi Arabia and is rich in historical morphological features and is vulnerable to new deformations such as Karsts formations and sinkholes. These deformations impact quality of life and stability of infrastructure and projects. The technical workflow involved the acquisition and preprocessing of Sentinel-1 radar images, followed by the application of DInSAR processing to generate displacement maps. These maps were analyzed using rigorous statistical methods, including time series analysis and spatial correlation techniques, to identify deformation trends and patterns. The findings reveal significant subsidence in the region, near Umm Jabal lake, exhibiting the highest subsidence rates. A linear regression analysis of the deformation data at this location yielded by a regression equation where the slope represents a steady subsidence rate of approximately 8.87 mm/year. This rate corresponds to a higher cumulative subsidence observed during the period from April 2024 to October 2024, highlighting the potential influence of seasonal aquifer discharge during the dry season. The aim of this study is to detect surface displacement and monitor their progress and evolution in order to provide early warning and signs for responsible entities to take needed safety measures and adjustments to avoid damage to life and projects infrastructure properties.

Keywords: Differential Interferometric Synthetic Aperture Radar (DInSAR), Generic Atmospheric Correction Online Service (GACOS), Surface Deformation Monitoring, Surface Displacements, Morphology, Single Look Complex (SLC), Polarization.

Introduction

Surface deformation has significant implications for infrastructure, natural resources, and environmental health, particularly in areas like Layla-Aflaj in central Saudi Arabia, where ongoing industrial activities and natural processes can cause ground movements. To monitor and understand these deformations, Differential Interferometric Synthetic Aperture Radar (DInSAR) has emerged as a valuable remote sensing technique, as introduced by Massonnet and Feigl [1]. By analyzing phase shifts in radar signals from repeated satellite observations, DInSAR provides insights into ground movements and surface displacements with millimeter-scale accuracy (Massonnet & Feigl, [1]. The technique's effectiveness is enhanced by parameters such as pixel spacing and adaptive phase filtering, with the latter notably discussed by Goldstein [2]. DInSAR has various applications, including earthquake studies, volcanic activity assessment, and urban subsidence research, as outlined by Burgmann et al. [3]. In the context of Layla-Aflaj, this study employs data from Sentinel-1 satellites, whose capabilities are detailed by Torres et al. [4], to analyze surface deformation over 16 months. The methodology involves selecting coherent InSAR pairs, processing the data, and quantifying subsidence, highlighting the technique's importance as reviewed by Hooper [5]. This approach underscores DInSAR's role in understanding regional subsidence implications, contributing valuable insights to environmental monitoring and urban planning.

Literature Review

There are no specific DInSAR studies available focusing exclusively on land deformation monitoring in Layla-Alaflaj area, which makes this study unique and first in the area. This research is expansion of work of the authors earlier published in IEEE IGARSS, Muhammad Asif et.al.[6] which explores first time implementing a satellite-based monitoring program using Sentinel-1 radar data from the European Space Agency's Copernicus project coupled with DInSAR methodologies covering an initial period of the first half of 2024 to the new span of study extended from January 2024 to April 2025. Nevertheless, DInSAR has proven to be effective in providing high-resolution insights into deformation patterns. With the advent of synthetic aperture radar (SAR) technology, particularly from the Sentinel-1 mission, the monitoring and analysis of surface deformation have significantly advanced. Various studies have harnessed Sentinel-1 data to track surface deformation phenomena. For instance, Pankratz, H.G., Sultan, M. et al. [7] utilized Sentinel-1 imagery to analyze environmental hazards associated with the rise of the Miocene Jazan salt diapir within Jazan city, Saudi Arabia, and identified areas at risk in its immediate surroundings and showcasing how interferometric synthetic aperture radar can detect millimeter-scale displacements over time. Similarly, Mohammadhossein Mohammadnia, Mahdi Najafi et al. [8] explored the potential of Sentinel-1 data for monitoring ground deformation using Interferometric Synthetic Aperture Radar (InSAR) technique with remote-sensing structural mapping to study the active surficial deformation of eight salt diapirs in the Kalut basin in central Iran. In the context of PInSAR, a study by Muhagir El Kamali, Ioannis Papoutsis et al. [9] employed Sentinel-1 data to monitor land surface deformations due to groundwater overexploitation in the agricultural area of Remah, UAE. Moreover, surface deformation studies have adopted advanced processing techniques such as machine learning to enhance the interpretation of Sentinel-1 data. For example, the work of Clayton M. J. Brengman et al. [10] integrated machine learning algorithms to improve the accuracy of deformation modeling, by developing SarNet, a CNN developed to detect, locate, and classify the presence of co-seismic-like surface deformation within an interferogram. Furthermore, researchers like Farhad Samadzadegana, Ahmad Toosi and Farzaneh Dadrass Javan et al [11] emphasized the synergy between multi-platform and sensor RS data fusion, encompassing visible-light images, multi/hyper-spectral images, RADAR images, LiDAR point clouds, thermal images, spectrometry samples, and geophysical data. This compilation of studies demonstrates the valuable contributions of Sentinel-1 data in advancing our understanding of surface deformation and innovation in data processing methodologies to fully harness the potential of this satellite technology.

Study Area

Central Saudi Arabia is a region of contrasts, featuring a mix of urban and rural areas, rocky desert landscapes, and rapid development in cities like Riyadh. The area is also significant for its proximity to oil fields and other natural resources, making it a socioeconomically important and environmentally sensitive zone. The Heet Formation, primarily composed of laminated anhydrite, plays a crucial role in the region's groundwater system, acting as an aquitard that traps fossil groundwater. This groundwater has been extensively used for domestic and agricultural purposes, but over-extraction has led to a significant decline in water levels, with some areas experiencing a drop of up to 100 meters Kempe & Dirks, 2008 [12]. The consequences of this depletion are evident in Ain Heet, the deepest cave in Saudi Arabia, where groundwater once flowed freely but has since receded. Furthermore, hypogene karstification in the region has resulted in the formation of sinkholes, particularly around Layla, where lakes were filled with water until the 1990s. The subsequent abstraction of groundwater caused these lakes to dry up, highlighting the dynamic interaction between geological formations and human activity. As the region continues to develop, monitoring surface deformation becomes essential to understand the complex relationships between human activities, geological processes, and environmental changes, Kempe & Dirks, 2008 [12].

Problem Statement

In the dynamic landscape of central Saudi Arabia, the pressing issue of land deformations poses significant threats to infrastructure stability and quality of life. The region, renowned for its historical morphological features, is increasingly vulnerable to emerging deformations such as Karst formations and sinkholes. These geological changes, often imperceptible in their early stages, can escalate unpredictably, leading to severe consequences, including damaged infrastructure and risks to human life. Traditional monitoring methods, while effective, often fall short in providing the proactive vigilance needed for such a vast and historically rich area. Enter Differential Interferometry Synthetic Aperture Radar (DInSAR), a cutting-edge remote sensing technique renowned for its cost-effectiveness and ability to monitor expansive geographical areas with high precision. DInSAR's prowess lies in detecting subtle terrain changes over time, making it an ideal solution for identifying gradual deformations that are often elusive to conventional methods. Differential Interferometry Synthetic Aperture Radar (DInSAR) has emerged as a powerful remote sensing technique capable of accurately detecting and monitoring surface displacements over vast regions. By leveraging radar data from the European Space Agency's Copernicus project, Sentinel-1 implementation of a satellite-based deformation monitoring study in Layla-Alaflaj as shown in Fig. 1. This study demonstrates the potential of DInSAR to provide quantitative insights into terrain deformations and to detect surface displacements, track their evolution, and provide timely warnings. By estimating surface deformation, the technology enables proactive measures to mitigate potential threats, safeguarding both human life and infrastructure. This innovative application of satellite-based radar interferometry not only provides critical insights for addressing the pressing geological concerns in Layla-Alaflaj but also establishes a vital framework for future interdisciplinary research in regions susceptible to land degradation and subsidence. By leveraging the advanced capabilities of InSAR to monitor ground deformation, this study aims to fill the gap between the detection of geological instability and the implementation of sustainable land management strategies. Ultimately, it offers a promising solution for regions confronting the insidious threats of land deformation, emphasizing the potential of remote sensing technologies to mitigate geological hazards and promote environmental resilience.

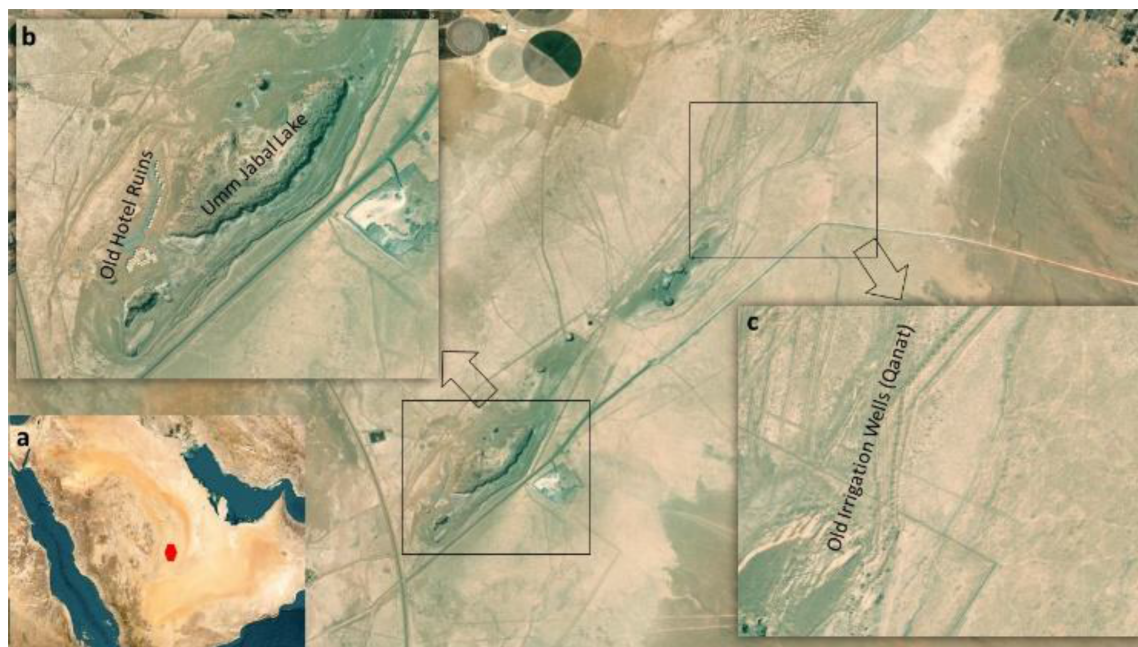


Figure 1—Study Area Map a) Study area location in Saudi Arabia. b) Remains of biggest Umm Jabal lake and a hotel for tourists. c) Remains of old irrigation wells (Qanat).

Methodology

Data Acquisition

The study leverages data from Sentinel-1, a key component of the European Space Agency's Copernicus program, which is renowned for its comprehensive Earth observation capabilities. Sentinel-1 is particularly distinguished by its utilization of C-band synthetic aperture radar (SAR), which is ideally suited for Interferometric SAR (InSAR) applications. The C-band radar offers several advantages, including its ability to penetrate vegetation and its sensitivity to surface changes, making it highly effective for monitoring terrain deformation and other geophysical phenomena. Satellite data were acquired for the period from January 2024 to April 2025, ensuring a comprehensive temporal coverage of the study area. Sentinel-1's 12-day revisit cycle provides a frequent and consistent stream of data, which is essential for capturing dynamic processes and changes over time. Furthermore, the satellite's ascending orbits are specifically designed to optimize observation geometry, minimizing geometric distortions and ensuring accurate measurements. The radar imagery was acquired in Single Look Complex (SLC) format, which preserves the full resolution and phase information of the raw SAR data. This format is crucial for advanced processing techniques, such as DInSAR, which relies on the precise phase information to detect subtle surface displacements. The Vertical-Vertical (VV) polarization configuration enhances the sensitivity to surface features, reducing noise and improving the overall quality of the data for the intended applications. The combination of high-resolution radar imagery, precise orbit control, and advanced data processing enables the study to achieve accurate and reliable results, making Sentinel-1 an indispensable tool for this research. This robust dataset, with its meticulous acquisition strategy, forms the foundation for a detailed and insightful analysis of surface deformation and related phenomena within the study area.

Selection of Coherent Pairs

The selection of suitable coherent pairs is a critical step in maximizing the effectiveness of Interferometric Synthetic Aperture Radar (InSAR) applications, particularly when utilizing data from the Sentinel-1 satellite. Sentinel-1 provides high-resolution C-band imagery with diverse temporal intervals, making it a valuable resource for various Earth observation studies. A coherent pair refers to two radar images that

exhibit sufficient temporal and spatial consistency to generate reliable interferograms. The reliability of these interferograms is directly influenced by two key factors: the perpendicular baseline, which is the spatial separation between the satellite orbits during the two acquisitions, and the temporal baseline, which is the time difference between the two acquisitions. To achieve optimal results, it is essential to select coherent pairs that minimize decorrelation effects. Decorrelation occurs when environmental factors such as atmospheric disturbances, vegetation growth, or other surface changes disrupt the consistency of the radar signals between the two acquisitions. These disruptions can significantly reduce the coherence of the interferogram, thereby limiting its reliability for precise measurements such as land deformation monitoring or soil moisture assessment. Zebker and Villasenor [13] emphasize the importance of careful pair selection to mitigate such effects and ensure high-quality interferometric results. In this study, the selection of coherent pairs was carried out through a systematic analysis of the temporal interval between ascending acquisitions. By identifying pairs with minimal time differences, the study aimed to reduce the likelihood of significant environmental changes between the two acquisitions. Additionally, amplitude information derived from the radar data was utilized to identify stable features in the scene. Stable features, such as urban structures or bare ground, are less prone to temporal changes and thus contribute to higher coherence levels. This approach also helps in avoiding interferograms with excessive noise, which can arise from highly dynamic environments such as densely vegetated areas or regions with significant surface modifications. The criteria for selecting coherent pairs in this study were defined by two primary parameters. First, a coherence threshold of 0.4 was established as the minimum acceptable coherence value for inclusion in the analysis. This threshold ensures that only pairs with sufficiently strong correlation between the two radar images are considered. Second, the pairs were required to fall within a specific temporal window, further minimizing the potential for environmental changes that could introduce decorrelation.

These criteria are consistent with methodologies outlined by Berardino et al. [14], who demonstrated the importance of rigorous pair selection in achieving accurate DInSAR results. The selection process resulted in the identification of coherent pairs, as illustrated in Figure 2. Each of these pairs was evaluated based on their temporal and perpendicular baselines, ensuring that they met the established criteria for coherence and reliability. The analysis of these pairs forms the foundation of the subsequent interferometric processing, enabling the generation of high-quality interferograms that are essential for accurate Earth surface characterization and change detection.

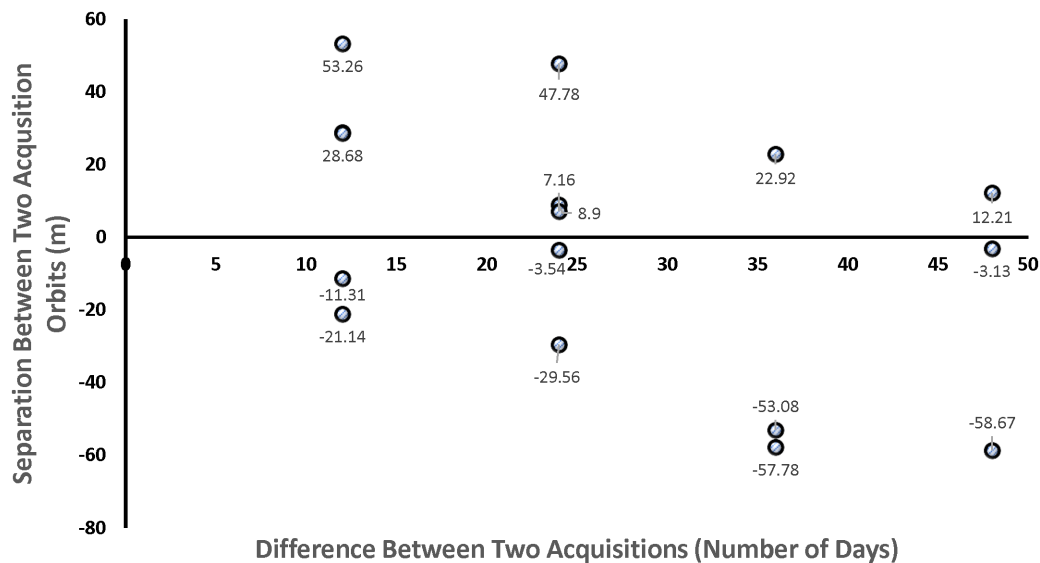


Figure 2—The diagram shows coherent pairs selection used for the current study, with its temporal and perpendicular baseline.

Data Preprocessing

Data preprocessing is a critical step in Differential Interferometric Synthetic Aperture Radar (DInSAR) that transforms raw radar data into meaningful information for analysis. Initially, raw data from satellites is not suitable for direct analysis, necessitating several preprocessing steps to enhance its usability. One key step was converting radar images from slant range to ground range. This conversion corrects the distance measured from the satellite to the target, adjusting it to represent the actual ground distance, which is essential for accurate spatial analysis and geolocation. Following this, co-registration of all Sentinel-1 image pairs to a master image was performed, ensuring precise alignment of each slave image with the master. The master image was selected based on low noise and high coherence, characteristics that ensure clarity and stability. This alignment process, achieved through techniques like cross-correlation, minimizes misalignments that could introduce errors in the interferogram. Proper co-registration reduces geometric distortions and phase errors, both crucial for accurate deformation measurements. Once these preprocessing steps were completed, the data was ready for interferogram generation, which is pivotal for subsequent DInSAR analysis. The foundation laid by preprocessing is vital, as any inaccuracies here can lead to unreliable results in the final deformation measurements. Thus, thorough preprocessing ensures the integrity and accuracy of the entire DInSAR process.

Interferogram Generation

Interferogram generation is a critical step in Differential Synthetic Aperture Radar Interferometry (DInSAR) that involves the complex convolution of two radar images to reveal phase differences related to surface displacement. In this research, interferograms were generated from coherent pairs of Sentinel radar images using the Sentinel Application Platform (SNAP) software. Once the images were co-registered and aligned, the interferogram was created by computing the pixel-wise complex multiplication of the master and slave images, followed by the conjugation of the slave image. This process highlights the phase differences between the two acquisitions, which are sensitive to changes in the Earth's surface. To enhance the quality of the interferograms and suppress noise, Goldstein phase filtering was applied. This adaptive filtering technique is widely used in DInSAR to reduce phase noise while preserving the coherence and fringe patterns that are critical for displacement measurements. The filtering process involved estimating the local phase gradient and using it to adaptively smooth the interferogram, thereby improving the signal-to-noise ratio. This step was particularly important for areas with low coherence, such as vegetated regions or areas with high temporal decorrelation [15].

Atmospheric Correction

Atmospheric correction is another critical step in Differential Synthetic Aperture Radar Interferometry (DInSAR) to ensure the accuracy and reliability of surface displacement measurements. Atmospheric effects, particularly tropospheric delays, can significantly compromise the precision of InSAR results by introducing spatial and temporal variations in radar signals. To address this, we implemented a robust atmospheric correction methodology in our past research, integrating advanced tools and techniques to mitigate these effects. Initially, Sentinel-1 InSAR data was preprocessed using the Sentinel Application Platform (SNAP) software to generate interferograms. SNAP was utilized to perform essential steps such as co-registration, interferogram formation, and phase filtering, which are foundational for subsequent analysis. Following the preprocessing, we employed the Generic Atmospheric Correction Online Service (GACOS) to account for atmospheric delays. GACOS leverages high-resolution numerical weather models to estimate atmospheric delay data, which was inputted into the system using metadata such as acquisition dates, satellite characteristics, and the geographical coordinates of the study area Yu, C., Li, Z., & Penna, N. T et al [16]. This service provided correction layers that were then formatted and integrated into the InSAR processing chain. By incorporating these corrections, we minimized the impact of atmospheric variations, enabling a more accurate interpretation of surface displacement measurements. This multi-step approach

not only enhanced the precision of our displacement measurements but also demonstrated the importance of combining advanced atmospheric models with rigorous validation protocols in InSAR studies. This methodology serves as a robust framework for future research, particularly in regions where atmospheric effects are significant, ensuring the production of reliable and accurate displacement maps.

Phase Unwrapping

In the context of satellite-based radar interferometry, the analysis of surface deformation over time is significantly enhanced by the application of Interferometric Synthetic Aperture Radar (InSAR) techniques. In this study after Interferogram generation and atmospheric correction we employed phase unwrapping process applied to all six Sentinel InSAR pairs using SNAPHU, by Jiang, Kun, Wenbin Xu and Lei Xie et al [17]. The phase unwrapping process was conducted using algorithms such as the Statistical-Cost Network-Flow Algorithm (SNAPHU). This algorithm is designed to minimize unwrapping errors and ensure a smooth transition of phase values across the image. The fundamental challenge of InSAR is to extract meaningful displacement information from the wrapped phase data, which can result in ambiguities due to the periodic nature of the phase measurements. Phase unwrapping technique are instrumental in reconstructing the continuous phase field from its wrapped form. Once phase unwrapping is achieved, the phase values are converted to displacement measurements utilizing the relationship by Richard Bamler and Philipp Hartl [18].

$$d = -\frac{\lambda}{4\pi} \Delta\phi$$

Where d represents displacement, λ is the radar wavelength, and $\Delta\phi$ is the phase difference between the two SLCs.

Vertical Displacement Calculation

Differential Synthetic Aperture Radar Interferometry (DInSAR), a powerful remote sensing technique, has been widely utilized to measure ground surface deformation with millimeter-level accuracy. The unwrapped phase, which represents the cumulative deformation signal over the observation period, was converted into vertical displacement maps where each pixel value corresponds to displacement in meters. To achieve this, the Line-of-Sight (LOS) deformation was estimated, assuming that horizontal displacement was negligible and that the majority of the movement was vertical. This assumption is often valid in scenarios where the deformation is primarily driven by subsurface processes, such as groundwater extraction or volcanic activity, which predominantly manifest as vertical motion. The displacement maps were initially visualized in grayscale, where brighter pixels indicated uplift and darker pixels represented subsidence. However, to enhance the interpretability and clarity of the results, a color-coding scheme was implemented. This approach involved applying a color ramp that differentiated uplift and subsidence using contrasting colors. The use of color-coding not only improved the visual distinction between areas of positive and negative displacement but also facilitated the identification of spatial patterns and gradients in the deformation field. Additionally, a color bar was included to provide a quantitative reference for the displacement values Massonnet, D., & Feigl, K. L. et.al [1]. The technical steps involved in this process were systematic and precise. The application of this methodology has proven particularly effective in monitoring natural and anthropogenic processes, such as volcanic activity, land subsidence, and mining-induced deformation. By providing high-resolution and accurate displacement measurements, DInSAR has become an indispensable tool for geodetic and geophysical studies. Furthermore, the integration of color-coding schemes has significantly improved the communication of results to stakeholders, facilitating informed decision-making in hazard assessment, resource management, and environmental monitoring. This research underscores the value of DInSAR in deriving vertical displacement maps and highlights the importance of careful data processing and visualization techniques in interpreting deformation patterns. Fig.3 shows the workflow followed for this study.

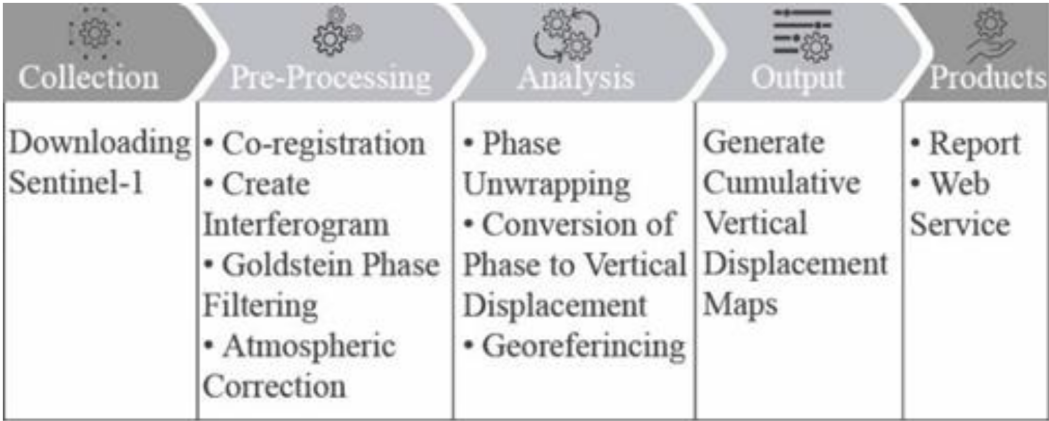


Figure 3—Methodology Workflow Diagram Summary

Results and Discussions

The study observed cumulative vertical displacement ranging from 1 to 14 mm over a sixteen-month period (January 2024 to April 2025). The spatial distribution of subsidence, as illustrated in Figure 4, revealed no patterns. The highest subsidence areas, represented by brown and orange colors, were concentrated along the southern and southwestern edges of the Area of Interest (AOI). Point "b" in Figure 4 recorded the maximum subsidence, while Point "a" corresponds to the location of the largest Umm Jabal lake. The brown color in the southeastern sand-covered area suggests localized subsidence patterns.

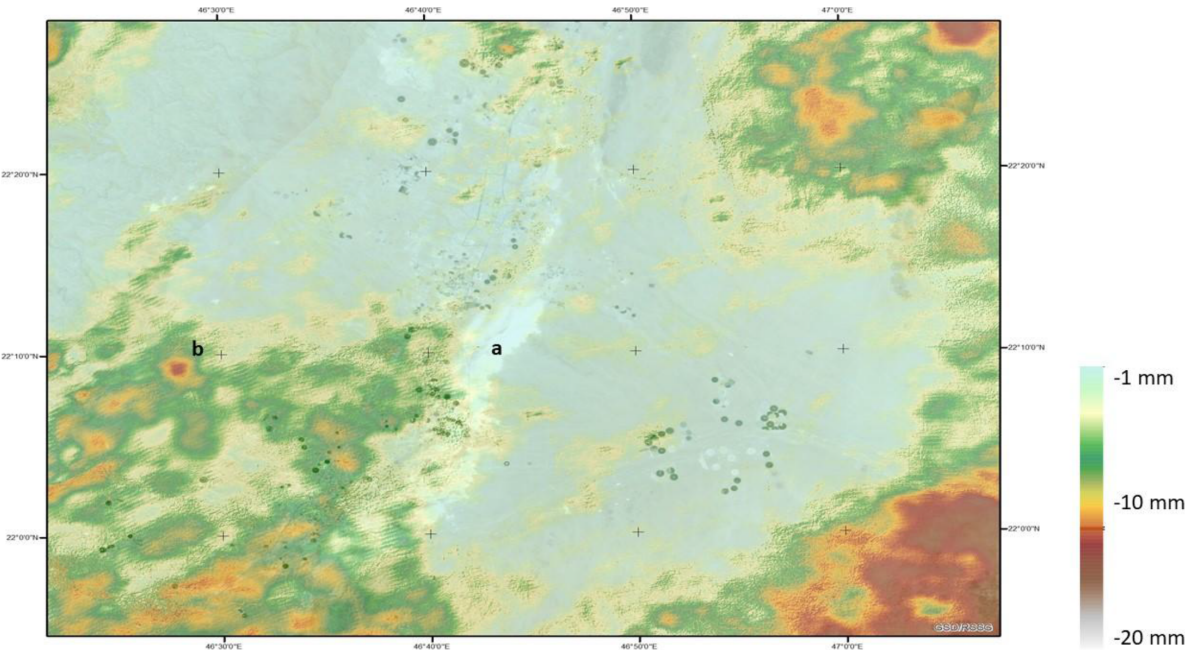


Figure 4—Subsidence map of study area overlaid on satellite image. Point a is location of biggest Umm Jabal lake and point b shows the highest subsidence observed.

Statistical analysis, including time series analysis and spatial correlation techniques, was conducted to identify deformation trends. However, no clear patterns were identified, likely due to the lack of prior studies in the region. A linear regression analysis was performed to quantify the overall deformation trend. The regression equation derived is:

Deformation = −2.43×10^{−5} Days −0.000194

where "Days" refers to the number of days elapsed since January 1, 2024. The slope of the regression line indicates a near-constant subsidence rate of approximately 8.87 mm/year. Figure 5 shows the temporal evolution of vertical ground deformation at Point "b" in Figure 4 as a steady, downward trend, with the most significant subsidence observed between April 2024 and October 2024, reaching a maximum displacement of 11.5 mm. The deformation pattern at this point exhibited a linear progression, suggesting a gradual and uniform process rather than episodic movement.

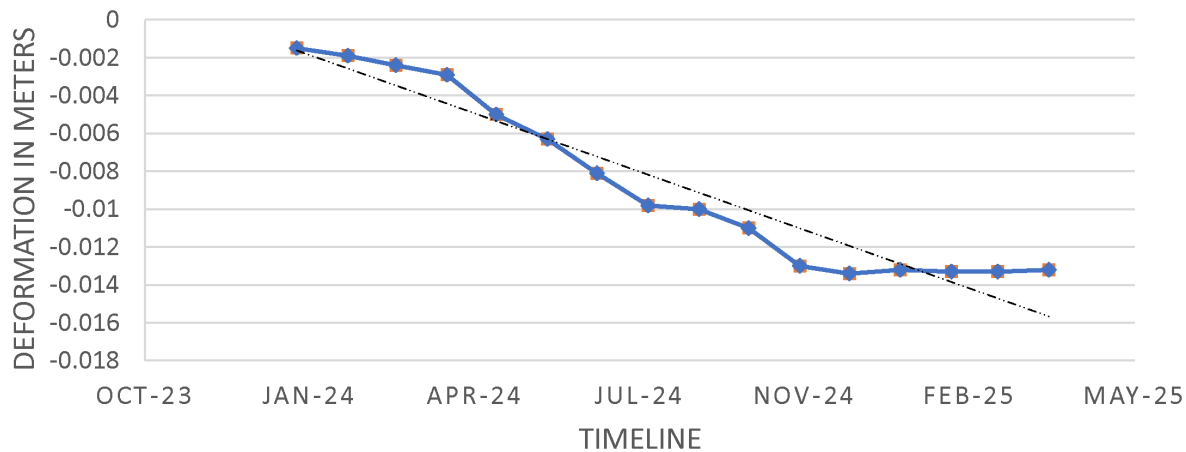


Figure 5—Temporal Evolution of vertical ground deformation, measured in meters at point b in Fig.4, over a monitoring period extending from January 2024 to April 2025

The observed subsidence patterns are likely influenced by a combination of geological and anthropogenic factors. The concentration of high subsidence in the southern and southwestern regions, particularly in karst terrains, can be attributed to the dissolution of soluble minerals and declining groundwater levels. Karst areas are inherently vulnerable to surface deformation due to the formation of voids in the subsurface, which can lead to subsidence.

The linear regression analysis indicates a steady subsidence rate, suggesting that the deformation process is primarily driven by consistent geological or anthropogenic activities, such as groundwater extraction or soil compression. The pronounced subsidence during the dry season (April to October) may be linked to increased aquifer discharge, highlighting the interplay between hydrological processes and surface deformation. The differential subsidence observed across the AOI underscores the heterogeneity of the region's geology and the localized impact of human activities. Areas with higher subsidence rates, such as Point "b," may face increased risks of structural instability and associated geohazards.

The findings of this study have significant implications for understanding and mitigating subsidence-related hazards in vulnerable regions. Subsidence phenomena, particularly in karst terrains and areas with intensive groundwater extraction, pose serious threats to infrastructure, ecosystems, and human settlements. The observed subsidence rates and patterns emphasize the need for continuous monitoring and further geotechnical analysis to identify underlying causes and predict future deformation behavior.

The integration of DInSAR-derived insights with in situ measurements is crucial for validating results and improving deformation modeling. This study highlights the utility of remote sensing techniques in monitoring surface deformation, particularly in areas with limited historical data. By leveraging advanced DInSAR techniques and robust statistical analyses, the research contributes to the broader goal of enhancing our understanding of the interplay between geological and anthropogenic factors driving subsidence.

In conclusion, the cumulative subsidence measurements (1 to 14 mm) underscore the significance of subsidence phenomena and their potential consequences. The study calls for further research to address the limitations of the current methodology and to explore the long-term implications of subsidence in vulnerable regions.

FUTURE WORKS

The findings of the current methodology underscore the potential for further enhancement through the integration of complementary datasets and advanced analytical techniques. Future research efforts should focus on incorporating additional data sources, such as GNSS monitoring meters, permanent reflectors, geological data, and seismic event records, to provide a more comprehensive understanding of ground deformation patterns. For instance, GNSS data can serve as ground truth to validate DInSAR-derived measurements, while permanent reflectors can enhance the coherence of radar signals, reducing artifacts and improving the accuracy of deformation estimates. Additionally, the integration of geological data, such as fault locations, lithology, and subsurface structures, can help correlate observed surface deformation with underlying geological processes, such as tectonic activity, subsidence, or underground mining. Seismic event data could further elucidate the temporal and spatial relationships between seismic activity and deformation patterns, enabling a more nuanced assessment of geodynamic processes.

To address the limitations of the current study, future work should prioritize the acquisition of higher-resolution C and X-band RADAR data (3-7 centimeters), which offers finer spatial and temporal resolution, enabling the detection of subtle deformation signals over longer monitoring periods. This would be particularly beneficial for monitoring urban areas, where small-scale movements in infrastructure (e.g., buildings, bridges) are critical for safety assessments. Furthermore, future studies could focus on calculating the rate of detected deformations, which is essential for understanding the progression and severity of ground movements over time. This would involve the application of time-series analysis techniques to DInSAR-derived displacement data, allowing for the quantification of deformation velocities and the identification of trends or anomalies.

The integration of DInSAR-derived measurements with geological assessments could significantly enhance hazard management strategies and urban planning initiatives. For example, deformation data could be used to identify areas prone to subsidence or landslide risks, informing the development of early warning systems and land-use policies. Moreover, the fusion of DInSAR results with other geospatial datasets, such as DEMs (Digital Elevation Models) and socioeconomic data, could provide a holistic framework for risk assessment and decision-making.

Looking ahead, the incorporation of Artificial Intelligence (AI) and machine learning techniques into the DInSAR workflow could revolutionize the detection and monitoring of ground deformations. These advanced algorithms could be applied to process large volumes of Geo-Bigdata, automating tasks such as noise reduction, image coregistration, and change detection. Machine learning models, such as neural networks and deep learning architectures, could also be trained to predict future deformation patterns based on historical data, enabling proactive hazard mitigation strategies. Additionally, AI-driven approaches could facilitate the integration of multi-source data, improving the robustness and reliability of deformation measurements.

In summary, the methodological framework developed in this study provides a solid foundation for advancing the application of DInSAR technology in deformation monitoring. By integrating additional datasets, refining analytical techniques, and leveraging AI and machine learning, future research could significantly enhance the accuracy, resolution, and applicability of DInSAR-derived deformation measurements, ultimately contributing to more effective hazard management and sustainable urban development.

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